BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

CPU Architecture, Registers and FDE

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: Answer all four before looking at the model answers.

1. Previous lesson: which storage has no moving read/write head?

2. Why might schools use an HDD for a large archive?

3. Why is RAM not permanent file storage?

4. Why choose an SSD for a portable laptop?

Today's objectives

Explain CU and ALU roles; trace a fetch using PC, MAR, MDR and ACC.

Key vocabulary - Read, cover, recall

PC
Address of next instruction

MAR
Address being accessed

MDR
Data or instruction transferred

ACC
Holds calculation results

Write one definition in your own words:

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Learn and practise

New learning - CPU units and registers

Control Unit (CU): coordinates and decodes instructions.

Arithmetic Logic Unit (ALU): arithmetic and logic.

PC	MAR	MDR	ACC
Next instruction	Memory address	Data / instruction	Result of work
Recipe bookmark	Shelf location	Ingredient fetched	Mixing bowl

I Do - Follow the fetch, decode, execute cycle

PC names the next address; MAR sends it to memory.

PC	MAR	MEMORY	MDR	CU
next address	address	instruction	transfer	decode

The PC is updated. The CU decodes the fetched instruction.

For an addition, the ALU calculates and ACC may hold the result.

Memory aid: PC = bookmark; MAR = shelf;

MDR = fetched ingredient; ACC = mixing bowl.

We Do - Explain each register

Use the model above. Complete each role in BLACK PEN.

PC tells us: _____

MAR stores: _____

MDR holds: _____

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Think and write

Code First - Predict > Write > Test > Improve

Start on paper. Check the model answers only on page 4.

```
a = 7   b = 5
print(a + b)
```

Predict the displayed output of the program.

Change b from 5 to 9. What is printed?

Which CPU unit calculates the sum?

You Do - Independent work

State what MAR and MDR each store. [2]

Explain the role of the Program Counter in fetch. [2]

What does the ALU do, and where may a result be held? [2]

Thinking: describe why the CU and ALU have different jobs.

Exit Ticket - True or False?

Answer before you turn to the green-pen page.

1. The Program Counter holds the next instruction address. ☐ T ☐ F
2. MDR holds the address of the next instruction. ☐ T ☐ F
3. The ALU carries out arithmetic and logical operations. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick correct answers. For a mistake, use the model answer then copy the FULL question and write a fresh answer below.

Do Now - Answers

Q1 Solid-state storage. ☐

Q2 High capacity at lower cost per GB. ☐

Q3 RAM is volatile. ☐

Q4 Fast access and resistance to knocks. ☐

We Do - Guided practice

PC: next instruction; MAR: address; MDR: data/instruction

Key vocabulary

PC ☐
Address of next instruction

MAR ☐
Address being accessed

MDR ☐
Data or instruction transferred

ACC ☐
Holds calculation results

Code First - Answers

Predict ☐
7 + 5 = 12

Write ☐
7 + 9 = 16

Improve ☐
ALU; the code is not a hardware simulation

Exit Ticket - Explanations

1. True

2. False: the PC holds that address

3. True

You Do - Self-mark

Q1	MAR: address; MDR: transferred data or instruction.	☐
Q2	PC holds the next instruction address, used for fetching.	☐
Q3	ALU performs arithmetic/logic; ACC can hold the result.	☐
Thinking	CU controls/decodes; ALU carries out calculations and logic.	☐

Try the question again

1. Copy the FULL question you got wrong (or the hardest):

2. In GREEN PEN, write a NEW answer. Show working: